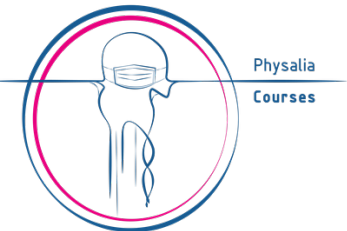
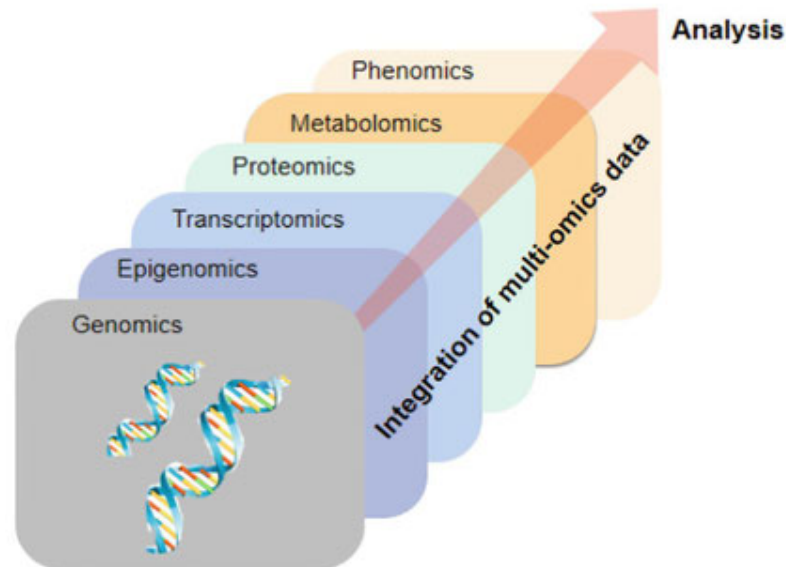


Machine Learning for Multi-Omics Integration

Physalia course, online, 9-11 June 2025

Course outline and practical information

Nikolay Oskolkov, Lund University, NBIS SciLifeLab



About us

Organizer: Carlo Pecoraro, Physalia courses

info@physalia-courses.org

Instructor:

Dr. Nikolay Oskolkov, Lund University, NBIS SciLifeLab

Nikolay.Oskolkov@biol.lu.se



@NikolayOskolkov



@osolkov.bsky.social



Personal homepage:
<https://nikolay-osolkov.com>

2007 PhD in theoretical physics

2011 medical genetics at Lund University

2016 working at NBIS SciLifeLab, Sweden

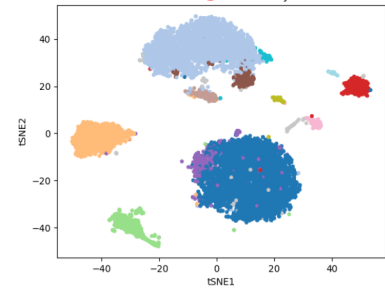
Lund University



SciLifeLab



Single cell



Biomedical data integration

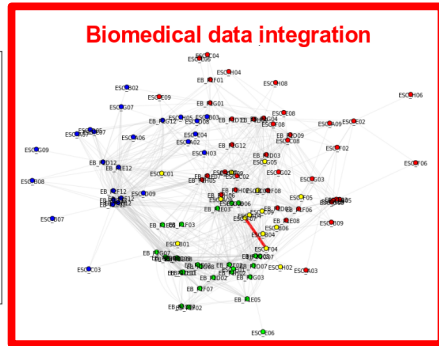
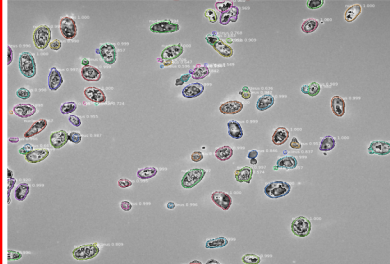
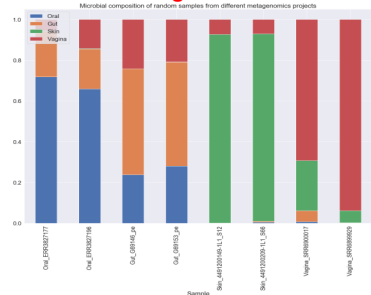


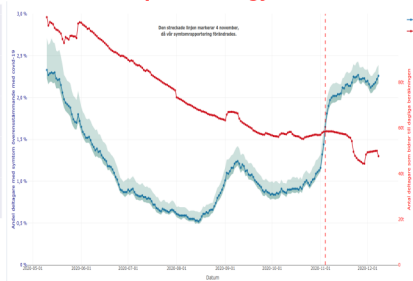
Image analysis



Metagenomics



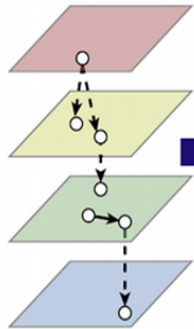
Epidemiology



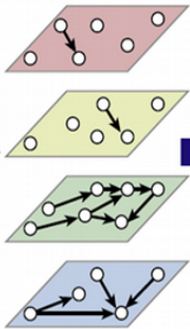
Ancient microbiome



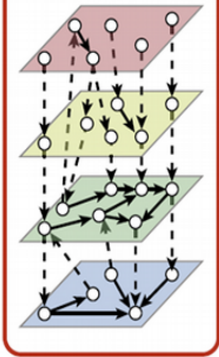
Conventional molecular biology



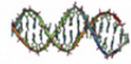
Single omics



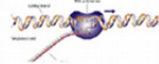
Trans-omics



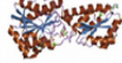
Genome



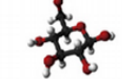
Transcriptome



Proteome



Metabolome



Measurement

NGS

RNA-seq (NGS)

Microarray

Mass spectrometry

Mass spectrometry

NMR

Yugi et al., Trends Biotechnol. 2016

scientific reports

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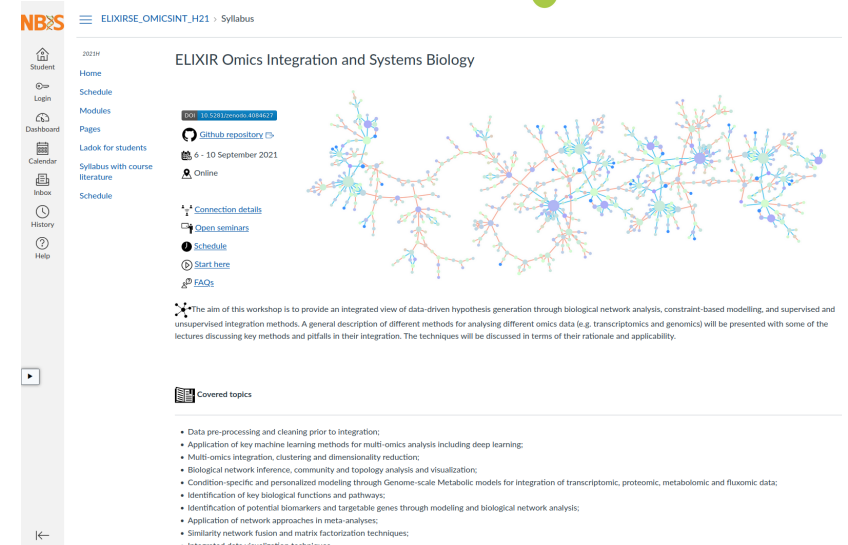
Article | [Open access](#) | Published: 25 June 2024

Predicting type 2 diabetes via machine learning integration of multiple omics from human pancreatic islets

[Tina Rönn](#), [Alexander Perfilyev](#), [Nikolay Oskolkov](#) & [Charlotte Ling](#) ✉

[Scientific Reports](#) **14**, Article number: 14637 (2024) | [Cite this article](#)

Rönn et al.,
Scientific Reports 2024



https://github.com/NBISweden/workshop_omics_integration

BOUGHT BY:

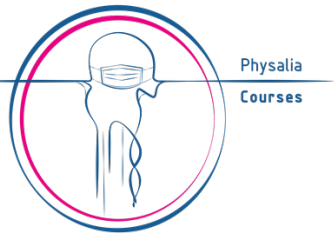


LEADING PROFESSIONAL SOCIETY FOR COMPUTATIONAL BIOLOGY AND BIOINFORMATICS
CONNECTING, TRAINING, EMPOWERING, WORLDWIDE

Last runs: February 2023 (online), 94 applications
October 2024 (onsite), 49 applications

About you

- Name
- University / Institute / Company
- Research interest(s)
- Previous experience with computational analysis and bioinformatics
- Motivation to join the course
- Expectations from the course



Course outline

Day 1: introduction of machine learning concepts for data integration, feature selection and supervised Omics integration

Day2: unsupervised Omics integration, factor analysis and graph intersection, neural networks for Omics integration

Day3: dimension reduction methods and UMAP for Omics integration, horizontal and vertical integration of single cell data



- To ask question please **raise your hand, unmute yourself and ask**. You can also ask questions in the zoom chat or slack channel for this seminar.
- **Please keep your camera on** as much as possible for better contact and communication.
- The course includes **6 lectures (~1h each) and 7 practicals (~1.5h each)**, there is a **15 min break** after each lecture and each practical
- During practicals, we will use **Rmarkdown and Jupyter notebooks**. At the end of each practical, I will use html-versions of the notebooks to go through command lines with my explanations.
- The material is based on data and problems from **computational biology**, however the concepts discussed are general and can be applied for other types of data.

Practical information: GitHub and Zoom

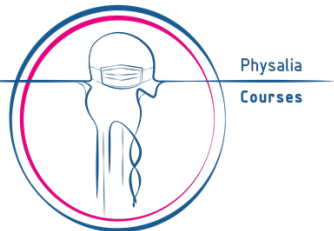
The course will take place in Zoom from 2 pm to 8 pm (CET, Berlin time)

Links to the Zoom room will be posted in the Slack channel

The course GitHub repository containing lectures and exercises is:

https://github.com/NikolayOskolkov/Physalia_MLOmicsIntegration_2025

Please bookmark this address!



Activities

Google Chrome

CONT x Refer x Mk Set up x Proje x B what x data x Deep x UMAF x R RStu x GitHu x Mach x GitHu x ChatC x css -H x Git-F x PDF f x +

Update

github.com/NikolayOskolkov/Physalia_MLOmicsIntegration_2025

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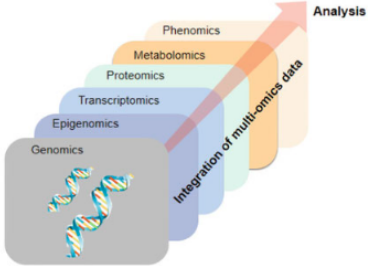
Go to file

<> Code

LeandroRitter added option to see rendered html 8b0c317 · 18 minutes ago 50 Commits

articles	added articles	2 days ago
practicals	added UMAP and single cell labs	yesterday
slides	added course logo	2 days ago
README.md	added option to see rendered html	18 minutes ago
command-line-basics.md	added schedule	2 days ago
course_logo.jpg	added course logo	2 days ago
data.zip	added data folder with practicals	2 days ago

README



Machine Learning for Multi-Omics Integration

Instructor

- Dr. Nikolay Oskolkov, Lund University, NBIS SciLifeLab

About

Physalia course Machine Learning for Multi-Omics Integration

Readme Activity 0 stars 1 watching 1 fork

Report repository

Releases

No releases published

Packages

No packages published

Contributors 2

- LeandroRitter Nikolay Oskolkov
- NikolayOskolkov Nikolay Oskolkov

Languages

- HTML 100.0%

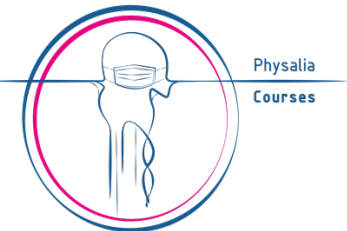
Practical information: Amazon Cloud (AWS EC2)



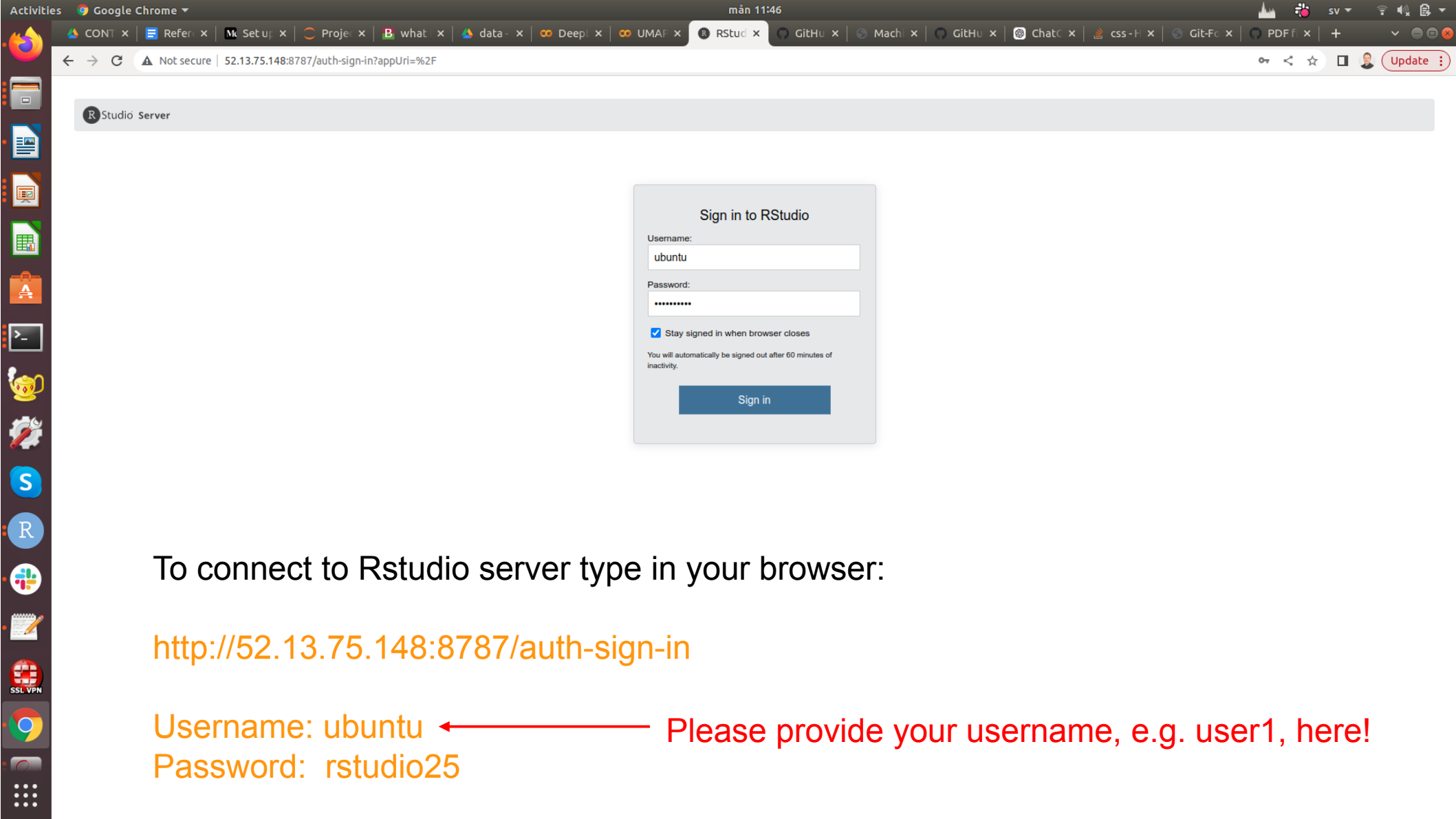
We will use the Cloud Computing service from Amazon, which we will access via `ssh` (secure shell protocol)

```
ssh -i multiomics.pem -X ubuntu@54.244.40.220
```

- The IP address changes every day
- Everyone is given a username, with a `home` and `shared` folders
 - List of usernames can be found in Slack
 - The `shared` folder is copy-only: do not delete, move, rename, or write



However, most of the time you will be using Rstudio server (for R exercises) and Google Colab (for Python exercises)!



To connect to Rstudio server type in your browser:

<http://52.13.75.148:8787/auth-sign-in>

Username: ubuntu ← Please provide your username, e.g. user1, here!
Password: rstudio25

Plan B (in case of network problems): install locally

```
if (!require("BiocManager", quietly = TRUE))  
  install.packages("BiocManager")
```

```
BiocManager::install("mixOmics")  
BiocManager::install("MOFA2")  
install.packages("rmarkdown")  
install.packages("knitr")  
install.packages("glmnet")  
install.packages("Rtsne")  
install.packages("uwot")
```

```
BiocManager::install(c('AnnotationFilter', 'BiocGenerics', 'GenomeInfoDb',  
  'GenomicFeatures', 'GenomicRanges', 'IRanges', 'Rsamtools', 'S4Vectors', 'TFBSTools',  
  'ggbio', 'motifmatchr', 'AnnotationDbi'))
```

```
install.packages("Seurat", dependencies = TRUE)  
install.packages("Signac", dependencies = TRUE)
```

To run Jupyter notebooks on Google Colab please do the following:

1. Clone this repo: https://github.com/NikolayOskolkov/Physalia_MLOmicsIntegration_2025
2. Unzip the data.zip folder locally and upload it to your Google drive
3. Start the Jupyter notebook from the Google drive and mount google drive:

#Mount Google Drive to Colab

```
from google.colab import drive  
drive.mount('/content/drive')
```

```
!ls -l /content/drive/My\ Drive/
```

```
import os  
os.chdir("/content/drive/My Drive/")
```

N.B. you need a Google account
to be able to run Google Colab!



Google Chrome

CONT x Refer x Mk Set up x Proje x B what x UMAF x DeepL x UMAF x R Stuc x GitHu x Mach x GitHu x ChatC x css -F x Git-Fc x PDF f x +

colab.research.google.com/drive/1Q57pB5IT92sRKQEuGERQSGdMVHmC6qZd#scrollTo=HLDfYRzqMzAe

Update

DeepLearningDataIntegration.ipynb

File Edit View Insert Runtime Tools Help All changes saved

+ Code + Text

RAM Disk Gemini

↑ ↓ ↻ ⌨ 📄 🗑 ⋮

Deep Learning for Data Integration

Biological and biomedical research has been tremendously benefiting last decade from the technological progress delivering DNA sequence (genomics), gene expression (transcriptomics), protein abundance (proteomics) and many other levels of biological information commonly referred to as OMICs. Despite individual OMICs layers are capable of answering many important biological questions, their combination and consequent synergistic effects from their complementarity promise new insights into behavior of biological systems such as cells, tissues and organisms. Therefore OMICs integration represents the contemporary challenge in Biology and Biomedicine.

[] from google.colab import drive
drive.mount('/content/drive')

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).

[] !ls -l /content/drive/My Drive/data/DeepLearningDataIntegration

total 49588
-rw-rw-r-- 1 root root 5105 Jan 10 12:26 CITEseq.py
-rw-rw-r-- 1 root root 2066933 Jan 10 12:26 DeepLearningDataIntegration.html
-rw-rw-r-- 1 root root 1747228 Jan 10 12:41 DeepLearningDataIntegration.ipynb
-rw-rw-r-- 1 root root 130233 Oct 3 2020 DeepLearningDataIntegration.jpg
-rw-rw-r-- 1 root root 5598515 Jan 10 12:26 DeepLearning_SingleCell_10X_1.3Mcells.html
-rw-rw-r-- 1 root root 3499 Jan 10 12:26 dim_reduct_CITEseq.py
-rw-rw-r-- 1 root root 3616 Jan 10 12:26 dim_reduct_scNMTseq.py
-rw-rw-r-- 1 root root 134229 Aug 8 2019 IntegrOmicsMethods.png
-rw-rw-r-- 1 root root 84 Jan 10 12:26 README.md
-rw-rw-r-- 1 root root 3772209 Jan 10 12:26 scATACseq.txt
-rw-rw-r-- 1 root root 4082712 Jan 10 12:26 scBSseq.txt
-rw-rw-r-- 1 root root 6141 Jan 10 12:26 scNMTseq.py
-rw-rw-r-- 1 root root 467570 Jan 10 12:26 scProteomics_CITEseq.txt
-rw-rw-r-- 1 root root 17128751 Jan 10 12:26 scRNAseq_CITEseq.txt
-rw-rw-r-- 1 root root 15547715 Jan 10 12:26 scRNAseq.txt
-rw-rw-r-- 1 root root 71008 Aug 8 2019 Strategies-for-multi-omics-profiling-of-single-cells-Three-major-types-of-molecules.png
-rw-rw-r-- 1 root root 3647 Jan 10 12:26 tsNE_on_Autoencoder_CITEseq.py
-rw-rw-r-- 1 root root 3635 Jan 10 12:26 tsne_on_autoencoder_scNMTseq.py

import os
os.chdir("/content/drive/My Drive/data/DeepLearningDataIntegration")

[] from IPython.display import Image
Image('DeepLearningDataIntegration.png', width=2000)

Connected to Python 3 Google Compute Engine backend